Benchmarking Non-Invasive Speech BCIs: Lessons Learned from the 2025 PNPL Competition

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Abstract

We present a retrospective on the 2025 PNPL Competition, an open machine-learning benchmark for decoding speech from non-invasive brain recordings, which ran from June to September 2025 and culminated in a workshop at NeurIPS that December. The competition used the LibriBrain dataset — over 50 hours of within-subject MEG with standard data splits — and challenged participants to tackle two tasks: Speech Detection (binary frame-level classification) and Phoneme Classification (39-class ARPAbet prediction). Across 155

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registered teams and 6,041 submissions, top F1-macro scores reached 95.6% for Speech Detection and 73.6% for Phoneme Classification, exceeding prior baselines of 68% and 44%, respectively, by wide margins. High-performing submissions converged on a common architectural template — temporal CNN or TCN backbones with task-specific recurrent or attention-based heads — whilst task-specific strategies such as temporal smoothing and ensemble voting proved critical for competitive performance. Hidden track evaluations suggest that main-track Speech Detection models generalised poorly to balanced class distributions. This could imply a reliance on class priors rather than robust neural decoding, highlighting a useful lesson: evaluation distributions should reflect intended deployment conditions. We further reflect on design choices, organisational lessons, and plans for future competitions in this series.

Keywords: brain decoding, speech detection, phoneme classification, MEG, benchmark

1. Introduction

By 2025, non-invasive speech decoding had reached an inflection point: deep learning methods were producing their first substantial results on MEG and EEG data, yet progress remained fragmented across incompatible datasets and splits, unreproducible models and training pipelines, and a field divided between neuroscientists and machine learning researchers. The stakes were not academic. Invasive brain-computer interfaces have demonstrated that decoding intended speech from neural signals is achievable (Moses et al., 2021) — with word error rates falling from around 20% in 2023 (Willett et al., 2023) to 2.5% in 2024 (Card et al., 2024). Brain surgery, however, remains a prohibitive barrier. A non-invasive route to speech decoding would transform the clinical picture, and MEG, with its combination of high temporal and spatial resolution, currently offers the best available signal to build a bridge from surgical systems to consumer-grade EEG devices. The primary challenge is that non-invasive decoding is substantially harder than invasive decoding; and the field has lacked the shared *infrastructure* needed to make rapid, reproducible progress.

This infrastructure gap inspired work to begin at Oxford in 2023, culminating in the first of multiple large-scale data releases that we are planning: LibriBrain (Özdogan et al., 2025), over 50 hours of within-subject MEG from a single participant listening to audiobooks, with standard train, validation, and test splits designed for reproducibility. LibriBrain's within-subject design was a deliberate choice, as deep single-subject data has been shown to yield better decoding performance per recording hour than shallow multi-subject data (d'Ascoli et al., 2025), and models trained on LibriBrain substantially outperform models trained on earlier datasets even at equivalent data volumes (Jayalath et al., 2025a). Isolated analyses using similar datasets existed before (Défossez et al., 2023; d'Ascoli et al., 2025; Jayalath et al., 2025b), but without standardised evaluation it has been difficult to know what generalisable lessons to draw. The precedent from computer vision here is instructive: shared benchmarks such as ImageNet (Russakovsky et al., 2015) and, more recently, the Natural Scenes Dataset in fMRI (Allen et al., 2022), have accelerated progress precisely by creating a common ground for comparison. In 2025, LibriBrain provided the foundation to do the same for non-invasive speech decoding.

The 2025 PNPL Competition (Landau et al., 2025b) was designed from the ground up with the LibriBrain data collection to build on this foundation, with accessibility as its guiding philosophy: bringing the machine learning community into an area that has histor-

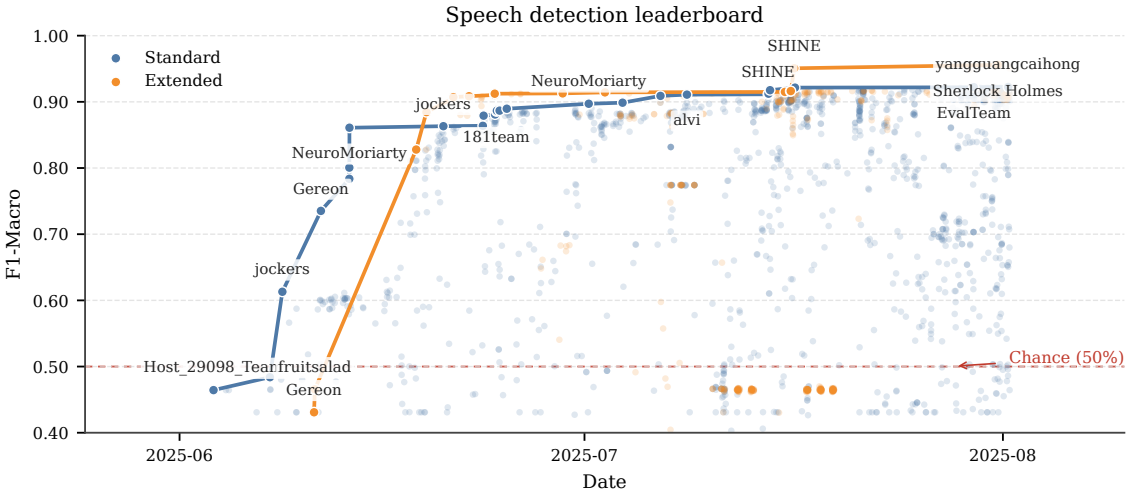


Figure 1: Leaderboard for Speech Detection (Phase 1). The x- and y-axes show submission dates and F1-macro scores (higher is better). Each circle is a submission, with blue and orange lines outlining improvements in the best models over time for the Standard and Extended Tracks. Balanced binary chance here is 50% (dashed red line). The final rankings, which used independent holdout data, are in Table 1; see Appendix A for the Phoneme Classification leaderboard.

ically demanded considerable domain expertise, by abstracting human neuroimaging data into standard tensor formats and providing end-to-end tooling. Across two phases and 155 registered teams, it produced 6,041 submissions on the benchmark tasks of Speech Detection and Phoneme Classification, culminating in a workshop at NeurIPS 2025 in San Diego. This paper is a retrospective on that competition. We synthesise patterns across competition submissions in Section 3.4, draw organisational lessons in Section 4, and conclude with plans for the 2026 PNPL Competition as the community advances towards word-level and ultimately full brain-to-text applications of non-invasive speech decoding.

2. Competition Overview

This section details the competition structure, tasks, evaluation metrics, and materials provided to participants. Complete specifications are available in the competition paper (Landau et al., 2025b) and on the competition website.[1]

2.1. Dataset and Splits

The competition used the LibriBrain dataset (Özdogan et al., 2025): over 50 hours of non-invasive MEG from a single participant listening to audiobooks, with time-locked speech/non-speech and phoneme annotations. Data were divided into train (51.57 hours), validation (0.36 hours), and test (0.38 hours) partitions, with additional holdout sessions reserved for (1) public leaderboard updates, (2) final rankings, and (3) hidden tasks (Table 2,

1. https://libribrain.com/

Section 3.4.4; also Appendix B). To reduce information leakage via slow features (Harris, 2021), all holdout sessions were acquired on different days from the training data.

2.2. Tasks

Two supervised decoding tasks were defined. **Speech Detection**: given $\mathbf{X} \in \mathbb{R}^{306\times T}$ (sensors $\times$ time points), predict for each $t \in \{1,\dots,T\}$ at 4 ms resolution whether speech is present ($y_t = 1$) or absent ($y_t = 0$). The general train, validation, test, and competition holdout data contained more speech than non-speech frames. **Phoneme Classification**: given a fixed window $\mathbf{X} \in \mathbb{R}^{306\times T'}$ (sensors $\times$ window size) aligned to a phoneme onset, predict which of 39 ARPAbet classes it corresponds to. Training and public evaluation data for this task used averages over 100 repetitions per phoneme to improve the signal-to-noise ratio (for decoding results with different amounts of averaging, see Özdogan et al., 2025).

2.3. Tracks and Rules

Each task was offered in a **Standard Track** (LibriBrain data only, emphasising algorithmic innovation) and an **Extended Track** (additional data and pre-training permitted), with separate leaderboards for each combination. Prize eligibility rules were updated between phases in response to Extended Track participation patterns; full details are given in Section 4.2.

The competition also included a **Hidden Track** for each task. These were not originally advertised, however this policy was updated during the competition (see Section 4.4). For speech detection, the hidden track used balanced speech and non-speech frames; for phoneme classification, single-trial (unaveraged) examples were used. See Section 3.4.4 for hidden track results, and Section 4.3 for a discussion of their implications.

2.4. Evaluation and Submission Protocol

All submissions were evaluated using the macro-averaged F1 score

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{F1}_k, \quad \mathrm{F1}_k = \frac{2\cdot \mathrm{TP}_k}{2\cdot \mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k},$$

where TP_k, FP_k, and FN_k denote true positives, false positives, and false negatives for each class k, with $K = 2$ for Speech Detection and $K = 39$ for Phoneme Classification. This metric was chosen over alternatives such as micro-averaged F1 (which weights classes by frequency) to penalise models that ignore minority classes, encouraging balanced performance across all categories. We generally report F1-macro scores as percentages.

Participants submitted predictions as CSV files via the EvalAI platform. Submissions were scored automatically against the public leaderboard holdout set, with results displayed on the competition website in real time. Final rankings were computed on an independent holdout set to control for overfitting on leaderboard feedback. For prize-eligible entries, Standard Track teams in prize positions were contacted to share training code to verify that no external datasets were used; Extended Track teams were required to confirm use of additional data beyond LibriBrain.

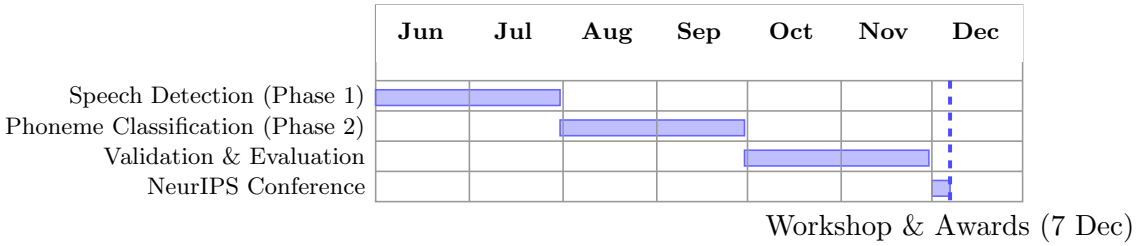


Figure 2: 2025 competition timeline: two competition phases, validation and evaluation, and the NeurIPS conference (2–7 December) with workshop/awards on 7 December.

2.5. Timeline

The competition was run over two phases as shown in Figure 2. Final verification was completed before the NeurIPS workshop on 7 December 2025.

2.6. Resources and Support

To lower barriers to entry, we provided: (1) a **Python library** (`pnpl`) with automatic dataset downloading and PyTorch-compatible data loaders; (2) **reference models** achieving F1-macro scores of 68% (Speech Detection) and 44% (Phoneme Classification); (3) four Google Colab **tutorial notebooks**,[2] executable with free GPU access, that provide code to train baseline models for both tasks, as well as examples for making competition submissions; (4) **blog posts** on problem context, baselines, and modelling approaches (Elvers et al., 2025; Landau et al., 2025a; Mantegna et al., 2025; Kwon et al., 2025); (5) a **Discord server** with 248 registered members; and (6) a **submission platform** with automatic evaluation and real-time leaderboard updates.[3] All materials remain publicly available.

3. Participant Outcomes

3.1. Participation Overview

The competition attracted 155 registered teams from at least 15 countries, with participants representing academia, industry, and independent researchers. Speech Detection (Phase 1) attracted 93 unique teams and 1,958 submissions; Phoneme Classification (Phase 2) engaged 62 teams and 4,083 submissions. We identified 7 teams that competed in both phases (see Appendix C for participation statistics).

Beyond competition submissions, community engagement with the supporting resources has been encouraging: the `pnpl` library was downloaded 14,736 times between release and March 2026 (Appendix D). Over the same period, the dataset was downloaded 15,086 times, and at its peak, the Discord server hosted 248 members. In total, 11 system-description papers (from 10 groups) were submitted to the NeurIPS 2025 PNPL Workshop by the following teams: BrainWatson (Suzuki et al., 2025), I_love_silksong (Li et al., 2026; Liang et al., 2026), neural2speech (de Zuazo et al., 2025), NTHU HMI LAB (Chien et al., 2025),

2. https://neural-processing-lab.github.io/2025-libribrain-competition/participate/
3. https://eval.ai/web/challenges/challenge-page/2504/overview

Task	Track	Score	Prize	USD	Winning Team	Paper
Speech Detection	Extended	95.6%	1st Extended	$1,200	Sherlock Holmes	
Speech Detection	Extended	95.1%	2nd Extended	$1,000	SHINE	
Speech Detection	Standard	92.18%	1st Standard	$1,200	Parameter Team	
Speech Detection	Standard	92.15%	2nd Standard	$1,000	TheLimbicLad	
Speech Detection	Standard	92.12%	3rd Standard	$800	alvi	
Speech Detection	Extended	90.74%	Special Prize	$300	I_love_silksong	
Speech Detection	Hidden	51.44%	Hidden Prize	$600	NeuroAI	
Phoneme Classification	Standard	73.6%	1st Standard	$1,200	neural2speech	
Phoneme Classification	Standard	73%	2nd Standard	$1,000	BrainWatson	
Phoneme Classification	Standard	65.7%	3rd Standard	$800	sigint	
Phoneme Classification	Extended	60.4%	Special Prize	$300	I_love_silksong	
Phoneme Classification	Hidden	5.38%	Hidden Prize	$600	September Labs	
			Total Prizes	**$10,000**		

Table 1: Winning teams and prize allocations. Scores reported as macro-averaged F1, sorted by task from best (highest) on competition holdout data. All prizes in USD. Where available, PDF icons in the 'Paper' column link to workshop papers.

Null-1 (2025), QuantumFiremen (2025), Saidnesh (2025), September Labs (Stepanov et al., 2025), Sherlock Holmes Team (2025), and SHINE (Xu et al., 2026). See Table 1 for rankings.

3.2. Track Participation and Rule Adaptation

The Standard Track consistently drew more participants than the Extended Track, with Standard-to-Extended ratios of 3.4:1 for Speech Detection and 3.1:1 for Phoneme Classification. Ultimately, the Extended Track did not attract sufficient competitive depth in Phase 1 to continue justifying ranked prizes in Phase 2, as we discuss in Section 4.2.

3.3. Performance Summary

Table 1 summarises all prize-winning submissions sorted by task and F1-macro score.

Speech Detection produced top scores exceeding 95% F1-macro in the Extended Track and 92% in the Standard Track, against a reference model baseline of 68% and balanced binary chance at 50%. The top three Standard Track submissions clustered between 92.12% and 92.18% — a range of just 0.06 percentage points — suggesting convergence on a performance ceiling within the constraints of the competition data. Extended Track submissions achieved higher absolute scores, with two teams — Sherlock Holmes (95.6%) and SHINE (95.1%) — demonstrating clear benefits from the additional resources, though one approach required a closed-set audio database and therefore has different applicability assumptions from open-vocabulary decoding (see Sherlock Holmes Team, 2025).

Phoneme Classification was substantially harder, consistent with its role as the more demanding task in the curriculum. The best Standard Track score was 73.6% F1-macro (neural2speech), well above the reference baseline (44%) and far above chance (2.6% for 39 balanced classes), but with room for further improvement. The gap between first and second place (0.6 percentage points) was narrower than between second and third place (7.3

percentage points), suggesting that neural2speech and BrainWatson converged on similar approaches whilst a larger gap separated these leaders from the rest of the field. Notably, Extended Track submissions did not outperform the Standard Track for Phoneme Classification: the best Extended score (60.4%) trailed all three Standard Track prize winners. This contrasts with Speech Detection, where the Extended Track submissions dominated.

3.4. Patterns Across Submitted Systems

The following analysis draws on prize-winning workshop papers, leaderboard trajectories, and structured information collected from high-performing teams. Because the number of machine learning systems available for analysis is small and many design choices were correlated across teams, the observations below should be read as empirical patterns and practical hypotheses rather than statistically established causal conclusions. We return to this caveat formally in Section 3.4.5.

3.4.1. A common architectural template

Despite the diversity of teams and backgrounds, high-performing submissions converged on a strikingly similar set of approaches. Most treated the MEG recording as a high-dimensional multichannel time series and built models accordingly, rather than adapting architectures from computer vision or natural language processing without modification.

The common template used the 306-channel MEG time series as input, applied a temporal convolutional backbone — typically using dilated convolutions or TCN-style blocks — and attached a task-specific prediction head. Dilated convolutions are well-suited to MEG: by increasing the dilation factor exponentially across layers, the receptive field grows to cover long temporal windows (8–30 s for Speech Detection) without a proportional increase in parameters. This architecture was used by the Parameter Team, SHINE, neural2speech, BrainWatson, and others, yielding a mean F1-macro of 87% across this group.

Within this shared template, task-specific routing diverged meaningfully. For Speech Detection, all three Extended Track prize-winning teams added bidirectional LSTM heads to capture long-range context in both temporal directions. For Phoneme Classification, the best-performing submission (neural2speech) used a Conformer-based head (Gulati et al., 2020), which combines self-attention with convolutional modules in a macaron-style architecture (FFN → attention → convolution → FFN), combining sensitivity to local spectral structure with broader contextual integration (de Zuazo et al., 2025). The second strongest Speech Detection submission (SHINE) used WaveNet-style layers as a variant on the same convolutional theme (Xu et al., 2026).

3.4.2. Task-specific drivers of performance

Whilst both tasks used the same LibriBrain recordings, they rewarded meaningfully different modelling choices.

Speech Detection. The largest single performance improvement came not from architecture but from post-processing. Model predictions at the 250 Hz MEG sampling rate oscillated between speech and non-speech labels every 4 ms — physiologically implausible,

given that speech segments last several seconds. Enforcing temporal coherence via smoothing yielded a mean F1-macro gain of approximately +21 percentage points across teams that applied it. Methods included median filters (window size $w = 40$), moving averages, and two-threshold hysteresis, all of which suppress high-frequency label switching whilst preserving genuine speech onset and offset transitions.

Phoneme Classification. The two largest performance drivers were ensemble voting and sample averaging. Combining predictions from multiple models trained with different random seeds, resolving disagreements by majority vote, produced the largest single gain of approximately +13 percentage points in F1-macro. All eight top-performing phoneme teams also exploited the 100-trial averaged data provided by the competition to improve the signal-to-noise ratio. A third factor was instance-level normalisation, which was critical for holdout generalisation in the winning team (neural2speech). A fourth factor was the utilisation of class imbalances across phoneme classes. Although F1-macro gives each class equal influence on the final evaluation score, standard example-averaged training losses give more influence to frequent phonemes. The top two submissions addressed this with inverse-square-root or temperature-scaled class weighting in the loss function.

3.4.3. Approaches that did not transfer

The clearest case of an approach that did not transfer was the use of pretrained vision backbones. Some teams reshaped the 306-channel MEG sensor array into image-like arrays compatible with architectures such as EfficientNet (Tan and Le, 2020). These approaches underperformed substantially, plausibly because the pretrained features were poorly matched to MEG waveform structure and image-like reshaping distorted the sensor array topology.

3.4.4. Evaluation robustness and hidden tracks

The hidden track evaluations tested generalisation beyond the training distributions. For balanced Speech Detection, the best score was 51.44% F1-macro — only slightly above balanced binary chance (50%) — despite the same models achieving above 90% on the holdout for the main tracks. For single-trial Phoneme Classification, the best hidden score was 5.38% F1-macro, above chance (2.6%) but well below the averaged-trial performance. This drop is expected given that single-trial MEG is a substantially harder problem than 100-trial averages (see analysis of averaging in Özdogan et al., 2025). These results admit several interpretations, which we discuss in the context of evaluation design in Section 4.3.

3.4.5. Statistical caveats

The number of high-performing submissions available for systematic comparison was small — approximately seven per track — which is insufficient for reliable inference about the independent contribution of any single design choice. Permutation tests across architectural features identified only one factor, the choice of Transformer architectures in Speech Detection, as approaching conventional significance ($p = 0.095$), and none of the features we explored survived Benjamini–Hochberg false discovery rate correction (all adjusted $p > 0.45$).

4. Organisational Outcomes

4.1. Competition Materials Lowered the Barrier to Entry

A central goal of the competition was to make non-invasive neural decoding accessible to AI researchers without prior neuroimaging experience. All the resources we provided — automatic data downloading, PyTorch-compatible data loaders, Google Colab notebooks with free GPU access, and blog posts covering problem context and modelling approaches — appear to have achieved this. The dataset was downloaded 1348 times in the first month and 155 teams registered, representing academia, industry, and independent researchers.

We set the reference models at a level that was deliberately more achievable rather than highly optimised, reasoning that an approachable baseline would encourage broader participation and deeper engagement. In practice engagement was indeed strong: the baseline was beaten by a large number of teams. Lessons from other competitions suggest that compute grants may also be as valuable as cash prizes for driving participation, particularly amongst student teams (e.g., Milani et al., 2020).

4.2. Even Small Friction Can Affect Participation

The two-track design balanced two goals. The Standard Track was intended to create a space where algorithmic innovation could compete on equal footing, without scale alone determining the winner. The Extended Track, by contrast, explicitly invited the bitter lesson (Sutton, 2019): teams were free to bring unlimited additional data, pre-training, and compute to see how far scaling could push performance. We assumed that teams competing on the Extended Track would be comfortable integrating external datasets into the `pnpl` library themselves, so the results surprised us.

In Phase 1, only two Extended Track submissions outperformed the Standard Track winners (Table 1), falling short of the three required to justify separate ranked prizes; a third group received a Special Prize rather than a ranked prize, so that future readers would not interpret limited Extended Track competition as evidence against scaling. By Phase 2, we had responded by replacing ranked Extended Track prizes with a single Special Prize, awarded to the best Extended Track effort without requiring it to outperform Standard Track entries.

Why did the bitter lesson not prevail and scaling dominate? One hypothesis is friction: because the Standard Track was comprehensively supported end-to-end, participants could focus entirely on modelling. By contrast, the Extended Track required integrating additional datasets into the `pnpl` library, and may have felt comparatively effortful. Compute resources were likely a second barrier. To address this, future competitions could explore compute partnerships and provide equivalent tooling for Extended Tracks, including ready-made loaders for external datasets. A growing number of efforts are emerging to support this (e.g., King et al., 2026), motivated in part by work on foundation models for the brain (e.g., Jiang et al., 2024; Yang et al., 2024; Csaky et al., 2024; Xiao et al., 2025; Huang et al., 2025; Cho et al., 2026; Jayalath and Parker Jones, 2026), so the 2025 PNPL Competition may simply have been ahead of the curve.

4.3. Evaluation Distributions Should Reflect Deployment

While the core evaluation methodology proved to be effective (e.g., automatic scoring via online platform using the macro-F1 score on independent leaderboard and final holdout splits), a lesson emerged from the hidden tracks. For Speech Detection, speech labels constituted approximately 78% of frames in the training, validation, test, and competition holdout data. This is a reasonable property for a dataset collected to study speech processing; given that MEG recording time is expensive, it makes sense to maximise the proportion of time in which a task stimulus is present. However, this meant that every evaluation split shared the same class imbalance, and there was no incentive not to exploit the empirical prior. As discussed in Section 3.4.4, the best Speech Detection models achieved above 90% F1-macro on the imbalanced holdout whilst barely exceeding chance on the balanced hidden evaluation. This is consistent with models learning to predict the dominant class (though we have not ruled out other explanations, such as atypically low signal quality in the balanced holdout session). Similarly, evaluating phoneme classification on averaged rather than single-trial samples meant that models were not required to be robust to individual sample noise.

These observations motivate a future design principle: evaluation distributions should reflect the intended deployment context, not the data collection process. In assistive communication, non-speech is likely to outweigh intended speech, and averaging across very large numbers of samples is likely infeasible. Future competitions would benefit from evaluating across a range of class distributions.

4.4. Disclose Hidden Tracks from the Start

As we elaborate in Appendix B, the nature of the hidden evaluation data was not initially disclosed — we had intended it as a surprise to reveal at the NeurIPS workshop. However, as results came in, several keen-eyed teams noticed that the statistical properties of the competition holdout appeared to shift and raised the alarm on the community Discord server. We refer to this episode jokingly as 'distribution gate'. Once community interest was sufficiently widespread we publicly confirmed the existence of a hidden evaluation track.

The engagement this generated was positive: teams were reasoning carefully about evaluation methodology and discussing together, both goals of the competition. It also motivates another lesson for future editions: tell participants at the outset if a competition holdout includes a hidden partition; that does not have to ruin the surprise of what it's for.

5. Conclusions

The 2025 PNPL Competition is an existence proof that open machine-learning benchmarking can drive rapid progress on non-invasive speech decoding. Starting from reference model baselines of 68% and 44% F1-macro, the machine learning community reached 95.6% and 73.6% across 155 registered teams and 6,041 submissions. Comprehensive materials and a task curriculum contributed to broad participation, and the competition generated organisational lessons for future editions. Together these advance the broader agenda: moving the curriculum forward to word-level and ultimately full text-level decoding, with evaluation conditions that reflect the needs of the people who will benefit from non-invasive speech BCIs.

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